

BetterSight: Immersive Vision Training for Basketball Players

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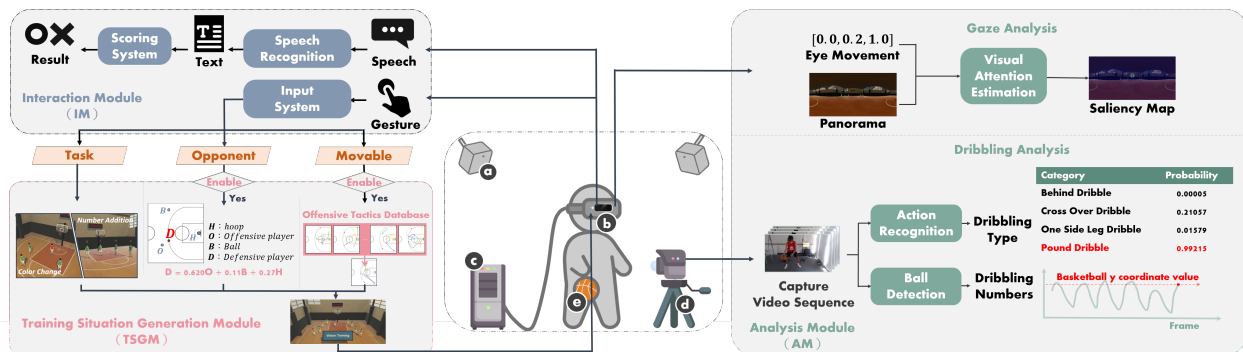


Figure 1: System framework of the proposed BetterSight. The devices we use contains (a) Lighthouse (b) StarVR headset with a microphone and hand tracking sensor (c) Server (d) Camera (e) Basketball.

ABSTRACT

Vision training is important for athletes and is key to win a sports game. Traditional vision training methods are suitable for sports that focus only on the ball. For basketball, however, players need to observe multiple moving objects (i.e., ball and players) concurrently on a large court. We propose BetterSight, an immersive vision training system for basketball that not only trains the vision of the player but also requires the player to dribble the ball stably, mimicking the situation in a real basketball game. BetterSight is composed of an Interaction Module (IM), a Training Content Generation Module (TCGM), and an Analysis Module (AM). IM allows the trainee to interact with the system more intuitively based on gesture and speech rather than the controller. TCGM simulates the training scenarios based on the training configurations selected by the trainee. AM collects the video sequences capturing the trainee and the trainee's eye movements during the training phase, and then analyzes the trainee's gaze, dribbling movements, and number of dribbles. The analyzed data can be used to evaluate the training effectiveness of using the proposed BetterSight.

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CCS CONCEPTS

• Computing methodologies → Artificial intelligence; Virtual reality.

KEYWORDS

vision training, virtual reality, sports training, dribbling analysis

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1 INTRODUCTION

With the advance of sports technology, trainers in various kinds of sports have started utilizing multimedia tools to improve the skill, cognition, and mindset of athletes. Vision training is a crucial skill in sports training, and many sports training methods for vision training have been proposed in the last decade [1]. Most existing methods present specially designed dynamic patterns through a 2D panel, requiring trainees to quickly discover and continuously track specific objects, which can improve the perception and concentration of players with regard to fast-moving objects. However, these methods are more suitable for sports that focus only on the ball, such as baseball and table tennis [5]. For multiplayer ball games,

players need to focus on multiple moving objects (i.e., ball and players) simultaneously on a large court. To the best of our knowledge, no systematic and useful approach has been proposed for vision training of team sports such as basketball.

To train the player's vision, most teams can only practice on a physical court, resulting in relatively high labor cost to simulate play scenarios involving teammates and opponents. To reduce the cost, we conducted interviews with basketball coaches to understand contemporary training scenarios and accordingly designed an intelligent system, BetterSight, to facilitate the vision training process. In a basketball game, an offensive player needs to pass the ball with quick and correct decision. Most of the time they need to make the decision while dribbling. Hence, during training, coaches would ask the player to not only perform specific dribbling actions but also complete the assigned simple tasks (e.g., accumulate the numbers performed by teammates' gestures) simultaneously when executing team sports offensive tactics. Inspired by the training scenario described above, we develop an immersive vision training system for basketball, which allows trainees to perform tactics in the first-person perspective in a virtual reality (VR) environment and complete the assigned tasks at the same time to enhance their vision ability. We further propose a dribbling analysis module to analyze the dribbling stability and dribbling movement, which can ensure that the trainee can keep performing the specified dribbling while completing the assigned vision training tasks.

2 SYSTEM DESIGN

Figure 1 illustrates our proposed immersive vision training system, BetterSight, composed of five devices and three computational modules. We utilize a StarVR head-mounted display (HMD) with 210° horizontal × 130° vertical field of view (FoV) to build the VR-based vision training environment. Compared to other HMDs that usually provide 110° FoV, the FoV of StarVR approximates to the FoV of human beings more closely. Lighthouses are used to detect the HMD and locate the position of the trainee. Since players in real basketball play do not handle anything but the ball, a microphone and an additional Leap Motion Sensor attached on the HMD are used to acquire trainee's voice and gesture instructions, respectively. The server receives the voice/gesture instructions and generates the corresponding virtual content. The rendered images and added sound effects are transmitted to the HMD via the connected cable. To analyze dribbling performance of the player while vision training, the trainee is asked to stand in front of an RGB camera performing dribbling movements assigned by the vision training system. More details about these modules are described as follows.

Interaction Module (IM). IM enables intuitive interactions between the trainee and the system. During training, the trainee is asked to hold a real basketball to perform dribbling movements, so it is more convenient for the trainee to interact with the system via hand gestures and speech than using handheld controllers. In this work, Leap Motion is used to track the hand position as well as to detect the hand skeleton, which enables trainees to directly set the training scenario configuration using hand gestures. When training begins, trainees must respond to the assigned tasks immediately. We apply the speech recognition technique provided by Microsoft Azure Cloud Cognitive Service to recognize the audio response,

and the recognition result is then sent to the proposed system to judge whether the answer of the assigned task is correct or not.

Training Content Generation Module (TCGM). TCGM is used to generate simulation scenarios for training. The module consists of Task Generation and Tactic Generation. Based on interviews with basketball coaches, we designed two kinds of tasks to achieve basketball vision training: a *number accumulation task* and a *color change task*. For the *number accumulation task*, our system generates and shows random numbers from 1 to 10 above the heads of virtual players when the trainee practices in VR. The trainee has to accumulate all these numbers and respond the answer as soon as possible. By achieving the *number accumulation task*, the trainee can train his/her vision by improving the ability of scanning the entire court to make quick decisions. For the *color change task*, the cloth color of a specific virtual player will be changed for a few seconds when the trainee practices in VR, and the trainee has to note this and instantly respond which number is above the color-changed virtual player. This task can improve the trainee's ability of finding a wide open opportunity. TCGM also provides different training scenarios by controlling the movements of virtual players, the presence or absence of virtual defensive players, and the tactics performed by the virtual players. We refer to the tactics used in NBA2K18 to record the offensive trajectory as our tactics database. Furthermore, experts are recruited to annotate the wide open opportunity for each tactic. We also apply Franks et al.'s method [3] to generate the corresponding defensive trajectory depending on the positions of offensive players, hoop, and basketball.

Analysis Module (AM). AM analyzes the trainee's gaze direction as well as dribbling performance during the training procedure. As StarVR HMD is equipped with eye-tracking sensors, we can obtain the trainee's eye movements in the virtual 3D space using the Tobii SDK. To visualize the eye movements, we apply equirectangular projection to convert the virtual environment into a panorama format and obtain the eye movements in 2D coordinates. Each 2D eye movement position can be represented by a mask of Gaussian distribution and accumulated along time to form a saliency map in a weighted manner, thereby visualizing eye movements during the training period. For dribbling analysis, we define four kinds of dribbling actions, i.e., behind dribble, cross-over dribble, one side leg dribble, and pound dribble. A camera is used to capture the trainee's external video sequence. TSM [4], an action recognition method, is utilized to train the model for classifying the dribbling actions. We also use Yolov4 [2] to detect the vertical movements of the basketball in the video sequences, forming the frequency peaks to count the number of times the trainee dribbles.

3 RESULT AND FUTURE WORK

We present BetterSight, an immersive vision training system for basketball players. Our demonstration video can be viewed at <https://youtu.be/EQ0i3dVtB9g>. Unlike existing vision training methods, BetterSight is suitable for multiplayer ball games such as basketball. In the future, we will focus on two main aspects: (1) Providing an interface for mobile devices (e.g., smartphones and tablets) so that coaches can externally set the training scenarios. (2) Adding additional mental training factors (e.g., noisy audience voices) to mimic a real basketball game.

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